Stochastic Finance and Risk Modelling

—

7 - American options

—

Original slides from Ioane Muni-Toke

Laurence Carassus and Gaoyue Guo

CentraleSupélec

Outline

First approach

Stopping time

Snell envelope and optimal stopping

Pricing and replication of American options

Outline

First approach

Stopping time

Snell envelope and optimal stopping

Pricing and replication of American options

Model and definition

- Consider a financial market consisting of $d = 1$ risky asset, defined on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ endowed with filtration $\mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T}$. For simplicity, we may assume $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_T = \mathcal{F}$.
- We focus on the pricing of American options. Recall that the holder can exercise the American optional at any time $t \in \{0, \ldots, T\}$. If the option is exercised at t, the holder will receive a payoff equal to H_t, where H_t is a positive v.a. that is $\mathcal{F}_t-$measurable. The option can only be exercised once.
- Example : the American put option of strike K on the risky asset is defined by $H_t = (K - S_t)^+$.
- We assume the market is arbitrage-free and complete. We denote by $\mathbb{Q}$ the equivalent martingale measure.

First approach

Proposition

Let U_t be the value of the American option at t. Then the sequence $(U_t)_{t=0,\dots,T}$ satisfies

$$\begin{cases} U_T = H_T \\ U_{t-1} = \max\left(H_{t-1}, S^0_{t-1}\mathbf{E}^{\mathbf{Q}}\left[\frac{U_t}{S^0_t}|\mathcal{F}_{t-1}\right]\right), \quad t = 1, \dots, T \end{cases}$$

- Dynamic programing formulation

Proposition

The discounted values $(\tilde{U}_t)_{t=0,\dots,T}$ of the American option is $\mathbb{Q}$-supermartingale. This is the smallest supermartingale dominating from above $(H_t)_{t=0,\dots,T}$.

Outline

First approach

Stopping time

Snell envelope and optimal stopping

Pricing and replication of American options

Definition and properties

Definition

A r.v. τ taking values in $\{0, \ldots, T\}$ is called $\mathbb{F}-$stopping time if $\{\tau \leq t\} \in \mathcal{F}_t$ holds for every $t \in \{0, \ldots, T\}$.

- Interpretation
- Example : no-anticipation for trading strategies

Proposition

(a) *Every constant $t \in \{0, \ldots, T\}$ is a stopping time.*

(b) *A r.v. τ is stopping time if and only if $\{\tau = t\} \in \mathcal{F}_t$ for every $t = 0, \ldots, T$.*

(c) *If τ_1 et τ_2 are stopping times, $\tau_1 \wedge \tau_2$ and $\tau_1 \vee \tau_2$ are both stopping times.*

Processus arrêté

Let τ be a stopping time and $(X_t)_{t\in\{0,\dots,T\}}$ be stochastic process.

Definition

We call $X^\tau = (X_{t\wedge\tau})_{0\le t\le T}$ a stopped process by τ, which is defined by for all $\omega \in \Omega$

$$X_{t\wedge\tau}(\omega) = X_{t\wedge\tau(\omega)}(\omega) = \begin{cases} X_t(\omega) & \text{if } t < \tau(\omega), \\ X_{\tau(\omega)}(\omega) & \text{if } t \ge \tau(\omega). \end{cases}$$

Proposition

(a) *If $(X_t)_t$ is adapted, $(X_{t\wedge\tau})_t$ is adapted.*

(b) *If $(X_t)_t$ is martingale, $(X_{t\wedge\tau})_t$ is martingale.*

(c) *If $(X_t)_t$ is submartingale, $(X_{t\wedge\tau})_t$ is submartingale.*

(d) *If $(X_t)_t$ is supermartingale, $(X_{t\wedge\tau})_t$ is supermartingale.*

Outline

First approach

Stopping time

Snell envelope and optimal stopping

Pricing and replication of American options

Snell envelope

- Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $(\mathcal{F}_n)_{n=0,\dots,N}$.
- Let $(Z_n)_{n=0,\dots,N}$ an $\mathbb{F}$-adapted positive process.

Definition

The Snell envelope of $(Z_n)_n$ is the process (E_n) defined by

$$\begin{cases} E_N = Z_N, \\ E_n = \max\left(Z_n, \mathbb{E}\left[E_{n+1}|\mathcal{F}_n\right]\right), \quad n = 0, \dots, N-1. \end{cases}$$

This is the smallest supermartingale dominating from above $(Z_n)_n$.

Proposition

Set $\nu_0 = \inf\{n \geq 0 : E_n = Z_n\}$. Then ν_0 is a stopping time and the stopped process $(E_{n\wedge\nu_0})_{n=0,\dots,N}$ is martingale.

Stopping time

Proposition

The stopping time ν_0 satisfies

$$E_0 = \mathbb{E}[Z_{\nu_0}|\mathcal{F}_0] = \sup_{\nu \in \boldsymbol{\nu}(0,N)} \mathbb{E}[Z_\nu|\mathcal{F}_0],$$

where $\boldsymbol{\nu}(0,N)$ is the collection of stopping times taking values in $\{0,\ldots,N\}$.

- ν_0 is said to be the optimal stopping time
- We can show

$$E_n = \sup_{\nu \in \boldsymbol{\nu}(n,N)} \mathbb{E}[Z_\nu|\mathcal{F}_0] = \mathbb{E}[Z_{\nu_n}|\mathcal{F}_0],$$

with $\nu_n := \inf\{k \geq n : E_k = Z_k\}$.

Outline

First approach

Stopping time

Snell envelope and optimal stopping

Pricing and replication of American options

American options

Proposition

The discounted values $(\tilde{U}_t)_{t=0,\dots,T}$ *defined by*

$$\begin{cases} \tilde{U}_T = \tilde{H}_T \\ \tilde{U}_{t-1} = \max\left(\tilde{H}_{t-1}, \mathbb{E}^{\mathbb{Q}}\left[\tilde{U}_t | \mathcal{F}_{t-1}\right]\right), \quad t = 1, \dots, T \end{cases}$$

is the solution to the optimal stopping problem as follows :

$$U_0 = \sup_{\tau \in \boldsymbol{\tau}} \mathbb{E}^{\mathbb{Q}}\left[\frac{H_\tau}{S_\tau^0}\right],$$

where $\boldsymbol{\tau}$ *denotes the collection of stopping times.*

American vs European

Proposition

Denote by V_t^a *the value of American option at* t *of payoff* $(H_0, \ldots, H_T)$*. Denote by* V_t^e *the value of European option at* t *of payoff* H_T*. We have*

(a) $V_t^a \geq V_t^e$ *for all* $t \in \{0, \ldots, T\}$ *;*

(b) *si* $H_t \leq V_t^e$ *for all* t*, then* $V_t^a = V_t^e$ *for all* t*.*

Corollary

For the case of call option, i.e. $H_u = (S_u - K)_+$ *and* $r \geq 0$*, we have* $V_t^a = V_t^e$ *for all* t*.*

- The last result is no longer true for the put option, if we add dividends, etc.

Replication of American options

Proposition

Let U_t be the value of American option at t of payoff $(H_0, \ldots, H_T)$. Assume that the market is complete. Then :

(i) *there exists a strategy (ξ_t^0, ξ_t) whose value process is V_t s.t. $V_t = U_t + A_t$, where A is a predictable and non-decreasing process with $A_0 = 0$;*

(ii) *On the behalf of seller, a perfect replication existe ;*

(iii) *On the behalf of buyer, the optimal stopping time τ is determined by $H_\tau = U_\tau$ and $\tau \leq \inf\{k \geq 0 : A_{k+1} > 0\}$;*

(iv) *A no optimal exercise time of buyer leads to a gain of seller whose adopts the perfect replication strategy.*